Rational Functions: Canceled-Factor Holes vs Vertical Asymptotes
Answer Key

Algebra 2 — Rational Functions

Quick Answers

1. $(x-3)(x+3)$, $(x-3)(x+4)$
2. $\frac{x+3}{x+4}$
3. $\frac{6}{7}$
4. -4
5. -2, $\frac{10}{7}$
6. $(x-2)^2$, $\frac{2x+4}{x-2}$
7. $\frac{5}{4}$, 2
8. —

Curriculum

Level: Algebra 2 — Rational Functions
HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

3. *If they got 0: used the original numerator over the reduced denominator, reporting the hole height as 0*
5. *If they got 10: stopped at the reduced numerator and reported 10 without dividing by the reduced denominator*
7. *If they got -5: stopped at the reduced numerator, reporting -5 instead of dividing by the reduced denominator*

1. Factor both parts of $f(x) = \dfrac{x^2-9}{x^2+x-12}$.

Solution: The numerator is a difference of two squares, so it splits into conjugate factors. The denominator is monic, so look for a factor pair of -12 that sums to $+1$: those are -3 and $+4$.

$$\boxed{x^2-9=(x-3)(x+3) \qquad \text{and} \qquad x^2+x-12=(x-3)(x+4)}$$

The shared factor is $x-3$; it is the only candidate for a hole.

2. Reduce $f(x) = \dfrac{x^2 - 9}{x^2 + x - 12}$ and state the lost restriction.
Solution: Divide out the shared factor:

$$f(x) = \frac{(x-3)(x+3)}{(x-3)(x+4)} = \frac{x+3}{x+4}$$

$$\boxed{\frac{x+3}{x+4}}$$

The reduced form is defined at $x = 3$, but f is not: cancelling removed the information that $x = 3$ was excluded from the domain. Write the restriction alongside the answer, $x \neq 3$ (and $x \neq -4$).

3. Height of the hole of f at $x = 3$.
Solution: A cancelled factor means the two-sided limit exists and equals the reduced form's value there, so evaluate the reduced form — not the original:

$$\lim_{x \to 3} \frac{x^2 - 9}{x^2 + x - 12} = \lim_{x \to 3} \frac{x+3}{x+4} = \frac{3+3}{3+4}$$

The hole sits at height $\boxed{\frac{6}{7}}$

Watch for: an answer of 0 comes from putting $x = 3$ into the *original* numerator ($3^2 - 9 = 0$) over the *reduced* denominator ($3 + 4 = 7$). Mixing the two forms is the single most common way to lose a hole height; reduce first, then substitute once.

4. Vertical asymptote of f.
Solution: After reduction $f(x) = \dfrac{x+3}{x+4}$, so the surviving denominator factor is $x + 4$. Set it to zero: $x + 4 = 0$. $\boxed{x = -4}$
Check the numerator there: $-4 + 3 = -1 \neq 0$, so the factor really does survive and the outputs grow without bound as $x \to -4$.

5. Find and fix: Nina's two asymptotes for $g(x) = \dfrac{x^2 - 25}{x^2 - 3x - 10}$.
Solution: Nina stopped at "the denominator is zero," which is only the domain question. Factor both parts before deciding:

$$g(x) = \frac{(x-5)(x+5)}{(x-5)(x+2)} = \frac{x+5}{x+2}$$

The factor she mishandled is $x - 5$: it cancels completely, so $x = 5$ is a hole, not an asymptote. Only $x + 2$ survives. Its zero is the asymptote, and the hole height is the reduced form at $x = 5$, namely $\dfrac{5+5}{5+2}$.

$$\boxed{\text{vertical asymptote } x = -2 \quad \text{and} \quad \text{hole at } \left(5,\ \tfrac{10}{7}\right)}$$

Watch for: an answer of 10 for the hole height stops at the reduced numerator and never divides by $5 + 2$.

6. Find and fix: Ray's hole for $h(x) = \dfrac{2x^2 - 8}{x^2 - 4x + 4}$.

Solution: Ray saw one $x - 2$ on each side and cancelled. The denominator is a perfect square, so it carries the factor *twice*:

$$\boxed{x^2 - 4x + 4 = (x - 2)^2}$$

The numerator carries it once: $2x^2 - 8 = 2(x - 2)(x + 2)$. Cancelling one copy leaves one behind:

$$h(x) = \frac{2(x - 2)(x + 2)}{(x - 2)^2} = \boxed{\frac{2x + 4}{x - 2}}$$

So $x = 2$ is a *vertical asymptote*, not a hole. The rule: a factor produces a hole only when it cancels completely; compare multiplicities, do not just look for a match.

7. Both at once: $k(x) = \dfrac{x^2 - x - 6}{x^2 - 4}$.

Solution: Factor first:

$$k(x) = \frac{(x - 3)(x + 2)}{(x - 2)(x + 2)} = \frac{x - 3}{x - 2}$$

The factor $x + 2$ cancels completely, so it produces the hole; the factor $x - 2$ survives, so it produces the asymptote. Height of the hole from the reduced form:

$$\lim_{x \to -2} k(x) = \frac{-2 - 3}{-2 - 2} = \frac{-5}{-4}$$

$$\boxed{\text{hole at } \left(-2,\ \tfrac{5}{4}\right) \quad \text{and} \quad \text{vertical asymptote } x = 2}$$

Watch for: an answer of -5 for the height is the reduced numerator alone; the division by -4 was skipped, and the sign of the result flips as a consequence.

8. Explain: is a zero denominator always an asymptote? *(Open response — flagged for manual review; a model answer follows.)*

Model answer: A zero denominator only guarantees that the function is undefined at that input; it does not say how the function behaves nearby, and behaviour near the point is what distinguishes the two cases. Compare the numerator's factors with the denominator's. If the offending factor also appears on top, at least as many times, it divides out entirely: the outputs approach an ordinary finite number from both sides, so the graph has a single missing point — a hole. If the factor survives, the denominator approaches zero while the numerator approaches something non-zero, so the quotient's size grows without bound and the graph has a vertical asymptote.

A table makes the contrast concrete. For $f(x) = \frac{x^2-9}{x^2+x-12}$ near $x = 3$: the outputs at 2.9, 2.99, 3.01, 3.1 all sit close to $\frac{6}{7}$ from both sides. Near $x = -4$: the outputs at -4.1 and -3.9 have opposite signs and enormous size. Same function, two different kinds of break.

Whether the function has a value at a hole: no. The point is missing from the graph, and $f(3)$ is undefined. Only the *limit* exists; a hole is a removable discontinuity because a value could be assigned, not because one already is.

What a reviewer should look for: the response must contrast a limit that exists with one that grows without bound, and must not claim the function is defined at the hole.